Software Engineering

Assignment-3

1) Write a program to find the following quantities for a C/C++ source code

   (a) Actual Program Length N
   (b) Estimated Program Length N_e
   (c) Program Volume V
   (d) Critical volume V_c
   (e) Program Level L
   (f) Program Effort E

Take as input the name of the file (.txt format) containing the C/C++ source code for which you have to compute the values of these 6 parameters.

Also find the invalid identifiers, constants and operators if any present in the given code. (using transition diagram).

FORMULAS : N = N1 + N2

$N_e = \eta_1 \log_2(\eta_1) + \eta_2 \log_2(\eta_2)$

$V = N \log_2(\eta_1 + \eta_2)$ bits

$V_c = (2 + \eta_2^*) \log_2(2 + \eta_2^*)$ bits

$L = V_c / V$

$E = V / L$ bits

where N1= Total number of all operators
N2= Total number of all operands
η_1 = Total number of distinct/unique operators
η_2 = Total number of distinct/unique operands
η_2^* = Distinct number of actual i/p operands

EXAMPLE :

```
//From “average.txt” file
#include<stdio.h>
void main()
{
   int a, b, c, avg;
   scanf("%d %d %d", &a, &b, &c);
   avg = (a + b + c) / 3;
   printf("avg = %d", avg);
}
```

The unique operators are:

#include, < >, void, main, (), {}, int, scanf, &, =, +, /, printf

The unique operands are:

stdio.h, a, b, c, avg, "%d %d %d", 3, "avg = %d"

a,b,c are i/p Operands. avg is o/p operand.

Delemeter : , ; ( )

Except identifiers, constants and delimeters, all entities are either operators or operands.